Token Identity as a Routing Signal for Residual MLP Experts

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Paper under double-blind review

Abstract

Learned mixture-of-experts routers use contextual hidden states to decide which parameters process each token. We ask how much of that routing signal can come from token identity alone when expert capacity is a small residual over a shared dense MLP. Our model maps each token ID to two narrow experts through a fixed, layer-specific lookup; the experts still transform the same contextual hidden state as the shared branch. In a matched single-seed comparison between 306.5M-parameter models trained on 8B FineWeb-Edu tokens, the token-routed model reaches diagnostic-stream NLL 2.9329 versus 2.9482 for dense, although the stream comes from the training split and routing is slower. Final checkpoints remain close on three zero-shot tasks. The routed model has lower WikiText-2 perplexity and C4-validation NLL, whereas dense has lower NLL on a Pile test subset. A learned-router diagnostic also changes ordering between short and 1.0003B-token budgets. The observed effect is therefore modest and corpus dependent; replication across seeds remains open.[1]

1 Introduction

Mixture-of-experts language models usually learn a router from each token's contextual hidden state (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024). The router can adapt expert selection to context, but it also introduces parameters, balancing objectives, and dispatch decisions. Fixed routing provides a useful counterpoint: if a simple token-level signal allocates parameters effectively, some benefits attributed to contextual routing may instead come from stable conditional capacity.

We test this idea in a deliberately asymmetric design. Every token passes through a shared dense SwiGLU MLP. Token identity selects two additional experts, each much narrower than the shared branch, and their output is added as a gated residual. Selection is context independent, but expert computation is not: both the shared branch and the selected experts receive the contextual hidden state produced by the transformer. The experiment therefore asks whether token identity is useful for selecting a residual parameter subspace, not whether a fixed token-ID lookup can replace contextual computation.

The primary comparison uses two 306.5M-parameter models trained for the same 8B-token budget. The models share their transformer backbone and stored MLP width; one uses a dense width of 4,096, while the other divides that width into a shared width of 3,840 and four routed experts of width 64, with two active per token. At the last common checkpoint of a fixed evaluation stream, the routed model reaches NLL 2.9329 and the dense model 2.9482. The routed model trains more slowly in the current implementation, and the evaluation stream comes from the FineWeb-Edu training split.

This study contributes a controlled observation rather than a general verdict on expert routing. It defines a shared-plus-residual architecture in which token identity controls only expert selection, verifies the realized routing and capacity accounting, and reports the full matched-token trajectory, standard tasks, and two independent-corpus NLL measurements. We also report a separate learned-router diagnostic with matched dense and fixed token-identity controls at short budget, followed by a 1B-token learned-router/dense comparison.

[1]Generative AI tools were used as general-purpose assistants for language editing, consistency checks between the manuscript and author-provided CSV/JSON metric files, and figure and revision scripting. The authors reviewed and verified all experimental results, numerical values, scientific claims, and final text, and take full responsibility for the submission.

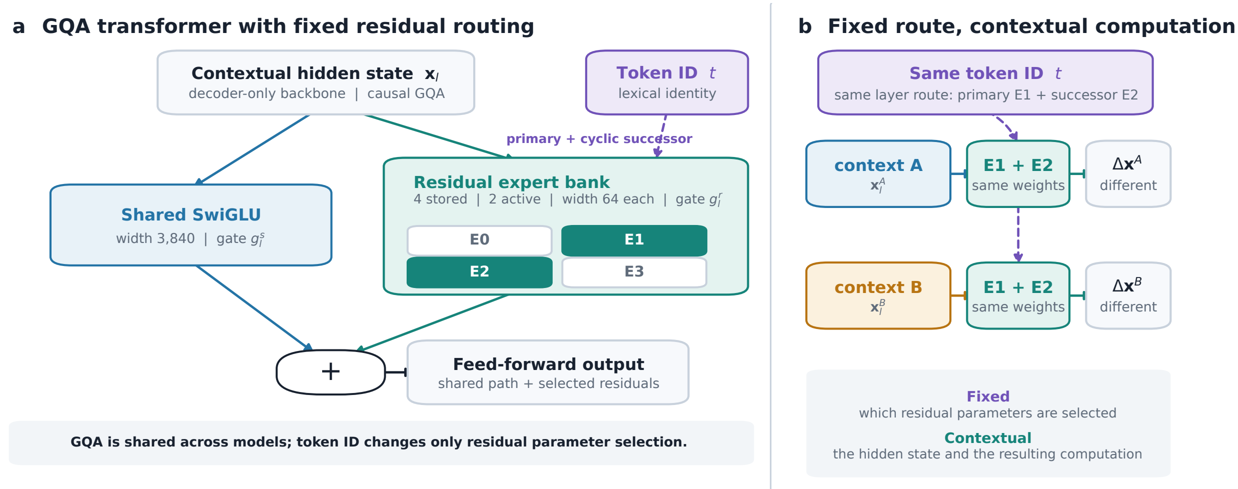

Figure 1: Token identity controls only the fixed top-2 lookup. The shared branch and selected residual experts receive the same contextual hidden state.

2 Related Work

Learned MoE routers use hidden states to select experts and commonly add an auxiliary objective or another balancing mechanism (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024). These systems test the value of content-adaptive specialization. Our experiment tests a different axis: whether stable token identity is already informative when routing controls only a narrow residual branch.

Hash Layers is the closest precedent (Roller et al., 2021). It showed that fixed hashes of local token features can compete with learned routing in language models. Our design keeps a dense MLP active for every token and applies fixed top-2 lookup only to a small residual capacity. This makes the routed path a correction to common computation rather than the main feed-forward transformation.

Shared experts also appear in architectures such as DeepSeekMoE (Dai et al., 2024), where they capture computation common to all tokens. Statistical analyses of shared-expert mixtures provide results under explicit estimation assumptions (Nguyen et al., 2025). We use the shared-plus-routed decomposition as an architectural prior; our measurements do not test those theoretical guarantees.

3 Architecture

3.1 Overview

The architecture changes only the feed-forward sublayer of a decoder-only transformer. Attention remains standard GQA with 16 query heads, four key/value heads, QK-Norm, RoPE, and causal scaled dot-product attention. Figure 1 isolates the changed computation. For token ID t and contextual hidden state $\mathbf{x}$, the token ID selects residual parameters while both MLP branches transform $\mathbf{x}$.

3.2 Fixed token-identity routing

Each layer stores a fixed top-2 lookup. The primary expert is a seeded permutation π_l of the token-ID residue:

$$r_{l,1}(t) = \pi_l(t \bmod E), \qquad E = 4. \tag{1}$$

The second expert is the cyclic successor of the primary:

$$r_{l,2}(t) = \big(r_{l,1}(t) + 1\big) \bmod E, \tag{2}$$

Because the 32,000-token vocabulary is divisible by four, each expert receives exactly 8,000 primary and 8,000 secondary token IDs. This balances vocabulary cardinality, not corpus traffic.

3.3 Shared MLP and residual experts

For contextual hidden state $\mathbf{x}$ and token ID t, the feed-forward output is

$$\mathrm{MLP}_l(\mathbf{x}, t) = g_l^s \,\mathrm{Shared}_l(\mathbf{x}) + g_l^r \sum_{j=1}^{2} 0.5 \;\mathrm{Expert}_{r_{l,j}(t)}(\mathbf{x}), \tag{3}$$

where g_l^s and g_l^r are learned scalar gates initialized to 0.5 and 0.5. The selected expert outputs use fixed weights 0.5/0.5. Every branch is a SwiGLU transformation,

$$\mathrm{Expert}_i(\mathbf{x}) = (\mathrm{SiLU}(\mathbf{x}\mathbf{W}^i_{gate}) \odot \mathbf{x}\mathbf{W}^i_{up})\mathbf{W}^i_{down} \tag{4}$$

and the shared branch has the same form with its own parameters. Token identity selects functions, but those functions operate on contextual states. The language-model objective is applied after the complete network; there is no independent expert objective.

3.4 Capacity and sparse dispatch

Let d be the model dimension, d_s the shared width, d_e the width of each routed expert, E the number of stored experts, and k the number selected per token. Ignoring biases, the routed MLP stores

$$P_{\mathrm{TR}} = 3d(d_s + Ed_e) \tag{5}$$

parameters per layer and evaluates intermediate width

$$d_{\mathrm{active}} = d_s + kd_e \tag{6}$$

per token. In the evaluated model, $(d_s, E, d_e, k) = (3840, 4, 64, 2)$: stored width is 4096, matching the dense baseline, and active width is 3968. The implementation groups tokens by their fixed assignments and evaluates only the selected routed experts. Small expert kernels and dispatch still add runtime overhead, measured in Section 5.1.

4 Experimental Design

4.1 Primary comparison

We train a token-identity residual model and a dense baseline, both with 306.5M parameters. They use the same 18-layer backbone, tokenizer, 2,048-token context, FineWeb-Edu stream, optimizer, learning-rate schedule, BF16 precision, and 8B-token budget. The global batch is 1,048,576 tokens (eight GPUs, batch 64 per GPU), so equal training steps imply equal tokens seen. The controlled difference is the MLP parameterization in Table 1.

4.2 Optimization

Both models use AdamW with $(\beta_1, \beta_2) = (0.9, 0.95)$, weight decay 0.1, gradient clipping at 1.0, a 5% warmup, and cosine decay to 10% of the peak learning rate. Residual output projections use

$$\sigma_{\mathrm{residual}} = \frac{0.02}{\sqrt{2 \times L}} \tag{7}$$

for L layers (Radford et al., 2019).

Table 1: Matched 300M model configurations.

Parameter	**Token-identity residual**	**Dense baseline**
Layers (L)	18	18
Dimension (d_{model})	1024	1024
Attention heads (n_h)	16	16
KV heads (n_{kv})	4 (GQA ratio 4:1)	4 (GQA ratio 4:1)
Routed intermediate dimension	256 total	—
Expert intermediate	64	—
Shared expert intermediate	3840	—
Experts ($n_{experts}$)	4, top-$k = 2$	1 (dense)
Shared/routed gate initialization	0.5 / 0.5	—
Vocabulary	32000	32000
Max context	2048	2048
Total parameters	**306.5M**	**306.5M**
Active MLP path/token	shared + 2 routed experts	dense SwiGLU

4.3 Routing and evaluation protocol

Checkpoint inspection shows that the stored primary lookup buffers in all 18 layers match Equation 1. The historical training code and exported evaluation implementation both derive Equation 2 from those buffers. The realized mechanism is therefore permuted-modulo primary routing with a cyclic successor, rather than a frequency-aware or learned assignment. The fixed evaluation stream is drawn from the FineWeb-Edu training split; it is consistent across the two runs but is not an independent held-out set.

4.4 Standard benchmark protocol

We export the final step-7,629 checkpoints to the same BF16 inference runtime and evaluate both without task-specific fine-tuning. EleutherAI LM Evaluation Harness v0.4.12 runs ARC-Easy, PIQA, HellaSwag, and WikiText-2 with no sample limit, zero-shot prompts, automatic batching, a maximum context of 2,048, and identical seeds. We report length-normalized accuracy for the three multiple-choice tasks and word perplexity for WikiText-2. A paired five-prompt generation check is run with CUDA Graph execution enabled before the full evaluation. The exact commands, checkpoint hashes, package versions, aggregate metrics, and runtime sanity record are included with the supplementary artifacts.

4.5 Independent-corpus protocol

We additionally score the first documents from pinned versions of English C4 validation and The Pile test split (**??**). Documents are joined with EOS and packed using the checkpoints' exact 32,000-entry tokenizer. For each corpus, both checkpoints score the same 128 contiguous blocks of 2,048 causal-LM targets (262,144 targets total) with Apple MLX/Metal; no weights are updated. C4 consumes 529 documents and The Pile 191 documents. These corpora are distinct from the declared FineWeb-Edu training stream, but no document-level decontamination is performed. Dataset revisions, artifact hashes, per-block values, and runtime versions are included in the supplement.

5 Results

5.1 Scaling Comparison (300M, 8B tokens)

The token-identity residual model begins behind the dense baseline and crosses it after roughly 0.8B tokens (Figure 2). At the last common evaluation checkpoint, step 7,500 or 7.864B tokens, its evaluation-stream

NLL is **2.9329** versus **2.9482** for dense, a difference of −0.0153. At the final matched training point, step 7,620, training NLL is **2.9231** versus **2.9324**. Table 2 labels the two metrics explicitly.

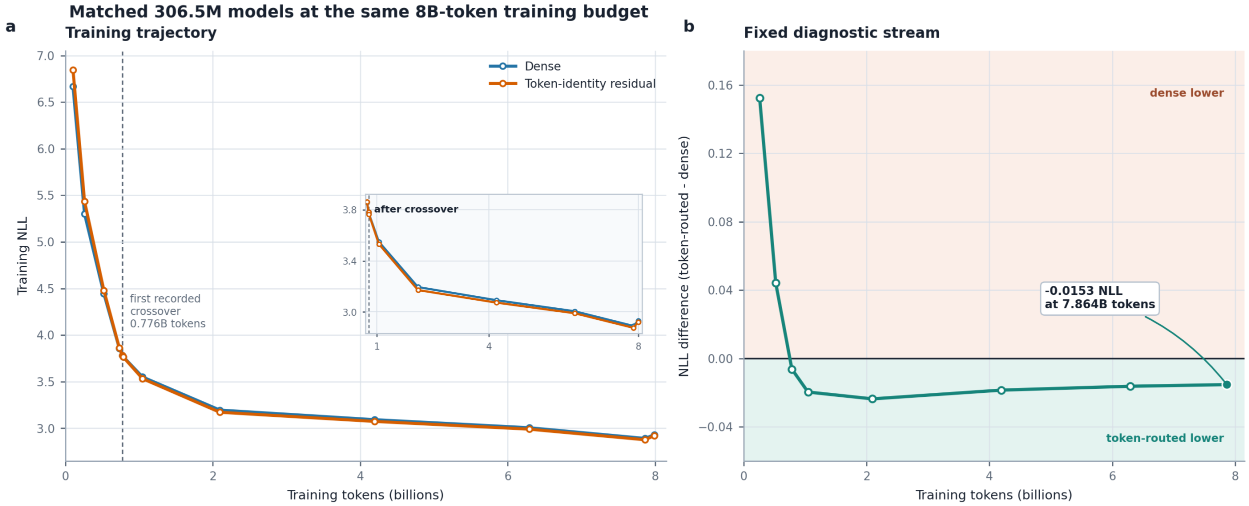

Figure 2: Matched-token loss measurements. Left: training NLL at recorded checkpoints. Right: token-identity residual minus dense NLL on the fixed FineWeb-Edu training-split evaluation stream; negative values favor the token-identity residual model. Markers are recorded checkpoints and lines guide the eye. The evaluation stream is not held out.

Table 2: Matched training and evaluation-stream NLL. The evaluation row uses the fixed FineWeb-Edu training-split stream; all other rows are training measurements. Negative differences favor the token-identity residual model.

Metric	**Step**	**Tokens**	**Dense**	**TR** ($k = 2$)	**Gap**
Train	100	105M	6.6698	6.8464	+0.177
Train	500	524M	4.4450	4.4796	+0.035
Train	1000	1.05B	3.5500	**3.5324**	**−0.018**
Train	2000	2.10B	3.1953	**3.1720**	**−0.023**
Train	4000	4.19B	3.0925	**3.0738**	**−0.019**
Train	6000	6.29B	3.0061	**2.9906**	**−0.015**
Eval. stream	7500	7.864B	2.9482	**2.9329**	**−0.0153**
Train	7620	7.99B	2.9324	**2.9231**	**−0.009**

Routing traffic. Final expert utilization is 0.2481/0.2635/0.2481/0.2403, with no dead expert. This is an observed property of the run, not a consequence of equal vocabulary cardinality.

Training throughput. On the same 8×B300 system and global batch, dense training reaches approximately 0.95M tokens/s and token-identity residual training approximately 0.75M tokens/s. The reported quality comparison is matched by tokens, not by elapsed time.

5.2 Zero-shot evaluation of the final 300M checkpoints

Table 3 reports the complete, untruncated benchmark runs. The token-routed checkpoint is 0.46 percentage points below dense on ARC-Easy, tied at the displayed precision on PIQA, 0.39 points above dense on HellaSwag, and has 0.59 lower WikiText-2 word perplexity. Each accuracy difference is smaller than one reported standard error, so this panel supports near-parity rather than a task-level superiority claim.

Table 3: Zero-shot evaluation of the final matched 300M checkpoints. ARC-Easy (2,376 examples), PIQA (1,838), and HellaSwag (10,042) report length-normalized accuracy in percent; WikiText-2 reports word perplexity (lower is better).

Model	ARC-Easy	PIQA	HellaSwag	WikiText-2 PPL
Dense	**49.07**	**63.49**	33.67	35.79
Token-identity residual	48.61	**63.49**	**34.06**	**35.20**

5.3 Independent-corpus NLL changes ordering

Table ?? reports the paired independent-corpus measurements. Token routing has lower NLL on C4 validation, whereas dense has lower NLL on The Pile subset. The opposite ordering constrains the result to a corpus-dependent effect rather than a universal generalization advantage.

Table 4: Independent-corpus evaluation of the final matched checkpoints. Each row uses 262,144 scored targets in 128 paired blocks. Δ is token-identity residual minus dense NLL; negative values favor routing. Confidence intervals use 10,000 paired bootstrap resamples over blocks.

Corpus	Dense NLL	TR NLL	Dense PPL	TR PPL	Δ NLL (95% CI)
C4 validation	3.4161	**3.4066**	30.45	**30.16**	$\mathbf{-0.0095}$ $[-0.0127, -0.0063]$
The Pile test	**2.8690**	2.8769	**17.62**	17.76	$+0.0079$ $[+0.0022, +0.0138]$

No decontamination against FineWeb-Edu was performed, so these measurements establish performance on independently selected corpora, not guaranteed document-level novelty.

5.4 100M Ablations

Panel A contains seven historical single-seed models of approximately 100M parameters, each trained for 1.0003B tokens on the recorded 2×B200 setup. The suite separates the shared branch, the residual branch, and several fixed top-2 assignments. Preserved configurations and launcher source show that the condition labelled frequency aware contains no populated frequency table and therefore realizes modulo-primary/balanced-secondary lookup. Nominal modulo and round-robin controls have equivalent primary assignments in this setup, so differences between those rows are descriptive rather than causal. The launcher intentionally disabled checkpoint retention and removed its temporary checkpoint directories; Panel A is therefore reported from raw metrics and must be rerun before it can receive new independent-corpus evaluation.

The train/evaluation step difference in the following panels reflects the recorded logging schedule, not a cross-checkpoint substitution. The 1B-token runs have a 954-step budget, log training metrics every 10 steps, and evaluate every 750 steps; step 950 is therefore their last common logged training point and step 750 their last common evaluation checkpoint. The short runs end at step 95, log training metrics every five steps, and evaluate every 75 steps, yielding the analogous step-95/step-75 pair.

Table 5: Panel A: historical matched-budget 100M ablations (2×B200, 1.0003B tokens). Evaluation loss uses the fixed FineWeb-Edu training-split stream at step 750. Throughput is comparable only within this panel.

Variant	Train @ 950	Eval. @ 750	Tok/s
Modulo-primary/balanced-secondary + shared	**4.8205**	**4.8887**	321k
Dense residual	4.8244	4.8958	397k
Modulo-adjacent top-2 + shared	4.8320	4.9016	321k
Round-robin top-2 + shared	4.8384	4.9072	321k
Random top-2 + shared	4.8875	4.9577	318k
Shared-only	4.9285	5.0012	**402k**
Modulo-primary/balanced-secondary, no shared	4.9693	5.0393	334k

The shared branch is important in this suite: removing it gives the highest evaluation loss, while using it without routed residual experts also trails the dense and shared-plus-routed variants. The balanced-secondary shared run records the lowest loss, narrowly ahead of dense. Because each row is a single seed and some nominal controls share the same primary assignment, these results support the component comparison but not a causal ranking of fixed assignment strategies.

To test the reviewer-relevant learned-router control before committing to full-budget runs, Panel B uses a separate o200k-tokenized FineWeb-Edu mmap shard and 99.6M training tokens on 2×RTX PRO 6000 Blackwell GPUs. All four variants use the same seed, token order, 98.2M-parameter backbone, batch, and schedule. The contextual routers select top-2 experts from hidden states and use no token-frequency statistics. The fixed token-identity control instead uses exact corpus-derived token counts to balance its fixed secondary route.

Table 6: Panel B: short-budget learned-router diagnostic (2×RTX PRO 6000 Blackwell, 99.6M tokens). Values are comparable only within this panel and not with Panel A. The reported train loss is language-model loss; the auxiliary term is excluded.

Variant	Params	Train @ 95	Eval. @ 75
Learned contextual top-2 + auxiliary balancing	98.213M	**7.4655**	**7.5647**
Dense residual	98.197M	7.4734	7.5766
Learned contextual top-2 + loss-free balancing	98.213M	7.5072	7.6052
Modulo-primary/corpus-balanced secondary + shared	98.197M	7.5310	7.6273

At this short budget, auxiliary balancing improves evaluation loss by 0.0119 over dense and 0.0626 over the fixed token-identity control. Its step-95 auxiliary term is 0.0101, no expert is dead, and its parameter excess over the fixed control is 15,360 parameters (0.0156%). Loss-free balancing remains behind dense. These single-seed early-budget differences motivated promoting only the auxiliary variant to the full token budget. Throughput is omitted because Panel A and Panel B use different hardware and the public fixed-routing path lacks the optimized sparse kernel used by the historical implementation.

The promoted run and its dense control both completed 1.0003B tokens on the same o200k setup as Panel B. Panel C compares this matched pair separately from Panel A.

Table 7: Panel C: matched full-budget comparison of the auxiliary-balanced learned router and dense control (2×RTX PRO 6000 Blackwell, 1.0003B tokens). This panel is not directly comparable with Panel A. The reported train loss excludes the auxiliary term.

Variant	Params	Train @ 950	Eval. @ 750	Tok/s
Learned contextual top-2 + auxiliary balancing	98.213M	4.7909	4.8580	95k
Dense residual	98.197M	**4.7849**	**4.8433**	**100k**

At step 750, expert traffic is 0.2400/0.2533/0.2610/0.2457 with no dead expert, and the auxiliary term is 0.0100. The dense control has 0.0147 lower evaluation loss and approximately 5.2% higher throughput. Thus the auxiliary router's 0.0119 short-budget advantage over dense does not persist to 1B tokens in this single-seed pair. This panel compares learned routing with dense computation; multi-seed replication remains open.

6 Discussion and Limitations

The result shows that a fixed token ID can select a useful residual parameter subspace in one matched run pair. It does not show that token IDs encode semantic expert classes. The assignments are modulo partitions up to a layer-specific permutation, and the selected experts transform contextual hidden states. A plausible interpretation is that stable token-conditioned capacity changes which functions are available to recurring lexical items while the shared MLP and attention preserve common contextual processing.

The experiment also does not isolate token identity from every alternative explanation. The 100M ablations in Section 5.3 include residual, random, and learned contextual controls. The short learned-router panel favors auxiliary balancing, but its matched dense control is ahead at 1B tokens while the promoted router completes without dead experts. Panels B and C use one seed and the same o200k tokenizer family as Panel A, but a different data shard and ingestion pipeline, implementation snapshot, and hardware. They therefore compare learned routing with dense computation on one setup without establishing that learned or fixed routing is generally superior. The historical fixed controls likewise cannot provide a causal ranking of assignment strategies because some nominal controls realize equivalent primary assignments.

The 300M comparison uses one seed per architecture, and its training-time loss-evaluation stream comes from the training split. The standard zero-shot panel is limited to three relatively simple multiple-choice tasks and WikiText-2; its accuracy gaps are smaller than one reported standard error. The independent C4 and Pile subsets each contain only 262,144 targets, were not decontaminated against FineWeb-Edu, and produce opposite model orderings. The routed implementation is also slower than dense at matched batch. Finally, the primary lookup realizes a permuted-modulo route with a cyclic successor, not frequency-aware assignment. These limits leave a modest architectural observation but constrain its scope.

7 Conclusion

Token identity provides a useful but distribution-dependent residual routing signal in the reported 306.5M-parameter run pair. The shared-plus-routed model finishes ahead on the FineWeb-Edu diagnostic stream, WikiText-2, and C4 validation, while dense is ahead on a Pile test subset and in training throughput. Because each architecture has one training seed, the study does not establish a general expected advantage. It instead poses a concrete question for learned MoE systems: how much contextual routing is necessary when a shared dense path already handles common computation?

Broader Impact Statement

This work studies an architectural modification for language models. The submission does not include model weights; any future release should follow the applicable data, safety, and licensing requirements.

Reproducibility Statement

The supplementary archive provides the PyTorch implementation, routing tests, realized run configurations, raw metrics, Apple MLX scorer, per-block independent-corpus results, and figure-generation scripts. It includes the 32k tokenizer for the primary experiment and the o200k configuration and shard-preparation script for the short learned-router diagnostic.

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